Lecture 3 · Classical Demand II

MWG Ch. 3.F–3.G · closing the square: Shephard, Roy, Slutsky

Advanced Microeconomics 1

Monday 28 September

Where we are

- Four objects exist and are linked by *level* identities: $e(p, v(p, w)) = w$, $h(p, u) = x(p, e(p, u))$, etc.
- Today we differentiate. Three derivative facts fall out in sequence:
  1. **Shephard's lemma**: $h = \nabla_p e$;
  2. **Roy's identity**: $x_\ell = -\dfrac{\partial v / \partial p_\ell}{\partial v / \partial w}$;
  3. **Slutsky equation**: $\dfrac{\partial h_\ell}{\partial p_k} = \dfrac{\partial x_\ell}{\partial p_k} + \dfrac{\partial x_\ell}{\partial w} x_k$.
- Then the payoff, in both goals' service: **(a)** acquires its test for the second data regime — the substitution matrix's property list *is* the rationality test when your data is an estimated demand function; **(b)** gets its tools — Shephard and Roy are exactly what Hausman (1981) differentiates to turn estimated demand into euros on Monday.

The envelope theorem

The only theorem you need today

Theorem · envelope theorem (constrained, differentiable case)

Let $V(q) = \max_x \{f(x,q) : g(x,q) = 0\}$ with Lagrangian $\mathcal{L}(x, \lambda, q) = f + \lambda g$, and let $x(q)$ be the (differentiable) maximizer. Then

$$\frac{\partial V}{\partial q} = \frac{\partial \mathcal{L}}{\partial q}\bigg|_{x=x(q)} = \frac{\partial f}{\partial q} + \lambda \frac{\partial g}{\partial q}.$$

- In words: to first order, ignore the re-optimization. The chain-rule terms through $x(q)$ vanish because the FOCs already set them to zero.
- Economic translation: when a price moves a little, the *behavioral response* has no first-order effect on the value — only the mechanical revaluation of the current choice does.
- Everything today, plus Hotelling (L6) and Shephard-for-cost (L7), is this theorem in different clothes.

Shephard and Roy

Proposition · MWG 3.G.1

For u continuous representing LNS, strictly convex $\succsim$: $e(\cdot, u)$ is differentiable and

$$h(p,u) = \nabla_p\, e(p,u), \qquad h_\ell(p,u) = \frac{\partial e(p,u)}{\partial p_\ell}.$$

Proof (envelope). $e(p,u) = \min_x \{px : u(x) \geq u\}$; the parameter p_ℓ enters the objective linearly with coefficient x_ℓ. Envelope: $\partial e / \partial p_\ell = h_\ell(p,u)$. $\square$

- Intuition: if p_ℓ rises by dp, the cheapest way to hold utility fixed costs $h_\ell\, dp$ more — *buy the same bundle*. Substituting away helps only at second order.
- Empirical gold: differentiate an estimated expenditure/cost function and you get demands, without ever solving a consumer problem. (Same move powers production-side econometrics, L7.)

The curvature dividend

Proposition · MWG 3.G.2

$D_p h(p,u) = D^2_{pp}\, e(p,u)$. Hence $D_p h$ is

- **symmetric** [Hessians are]
- **negative semidefinite** [e concave in p]
- **singular**: $D_p h(p,u)\, p = 0$ [h h.d.0 in p + Euler's formula]

- This is why we cared that e is concave: differentiate Shephard once and the entire structure of substitution falls out of *one curvature fact*.
- NSD diagonal: $\partial h_\ell / \partial p_\ell \leq 0$ — the compensated law of demand again, now infinitesimally.

Proposition · MWG 3.G.4

At interior solutions with v differentiable and $\partial v/\partial w > 0$:

$$x_\ell(p, w) = -\frac{\partial v(p, w)/\partial p_\ell}{\partial v(p, w)/\partial w}.$$

Proof. Differentiate the identity $v\big(p,\ e(p, u)\big) = u$ with respect to p_ℓ:

$$\frac{\partial v}{\partial p_\ell} + \frac{\partial v}{\partial w} \cdot \underbrace{\frac{\partial e}{\partial p_\ell}}_{= h_\ell \ \text{(Shephard)}} = 0 \implies h_\ell = -\frac{\partial v/\partial p_\ell}{\partial v/\partial w},$$

then evaluate at $u = v(p, w)$ so $h_\ell = x_\ell$. □

- Reading: a price increase hurts in proportion to how much you buy ($\partial v/\partial p_\ell = -\lambda x_\ell$), converted into consumption units by the marginal utility of wealth.

The Slutsky equation

The derivation (know it cold)

Start from the identity linking the two demands: $h_\ell(p,u) = x_\ell\big(p,\ e(p,u)\big)$.
Differentiate with respect to p_k:

$$\frac{\partial h_\ell}{\partial p_k} = \frac{\partial x_\ell}{\partial p_k} + \frac{\partial x_\ell}{\partial w} \cdot \frac{\partial e}{\partial p_k} = \frac{\partial x_\ell}{\partial p_k} + \frac{\partial x_\ell}{\partial w} h_k(p,u).$$

Evaluate at $u = v(p,w)$, where $h_k = x_k(p,w)$:

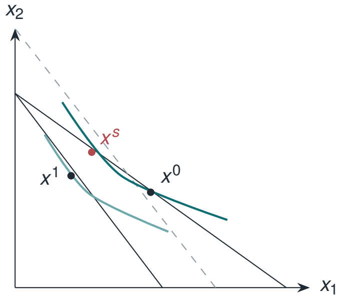
Slutsky equation (MWG 3.G.3)

$$\underbrace{\frac{\partial h_\ell}{\partial p_k}}_{\text{substitution}} = \underbrace{\frac{\partial x_\ell}{\partial p_k}}_{\text{total (observable)}} + \underbrace{\frac{\partial x_\ell}{\partial w} x_k}_{\text{income effect}} \equiv s_{\ell k}(p,w).$$

- Same S matrix as Lecture 1's choice-based construction — but now it *equals* $D_p h = D^2_{pp} e$, and inherits that object's full structure.

Anatomy of a price change

- Price of good 1 rises: budget pivots in. Decompose the move $x^0 \to x^1$:
  - **substitution** ($x^0 \to x^s$): slide along the indifference curve to the new relative price;
  - **income** ($x^s \to x^1$): shift in to the actual budget.
- Substitution is always nonpositive (own-price). Sign of the total is a horse race.
- **Giffen anatomy**: inferior good + large budget share $\Rightarrow$ income effect can win. Consistent with Jensen–Miller's rice evidence: staple, poor households.



Goal (a), second data regime: testing a demand *function*

	finite data $\{(p^t, x^t)\}_{t=1}^{25}$	a demand function $x(p, w)$
test for rationality	**GARP** (your loops, Thursday)	S **symmetric, NSD**, $Sp = 0$
equivalence theorem	Afriat (Monday)	integrability (Monday)
where it bites	lab, panels, this room	any estimated AIDS/QUAIDS syster

- Same question — *could a coherent preference have generated this?* — posed to two kinds of data. You cannot run GARP on a continuum; the matrix restrictions are GARP's translation.
- Why is symmetry the rationality condition? The **price-loop money pump**: with $s_{12} \neq s_{21}$, walk prices around a closed loop, compensating at each step; the compensations fail to cancel and the consumer bleeds money *every lap*. Symmetry $=$ path independence in price space $=$ transitivity. Same disease as the finite-data cycle, different data.

The substitution matrix: complete property list

Proposition · properties of $S(p, w)$ under preference maximization

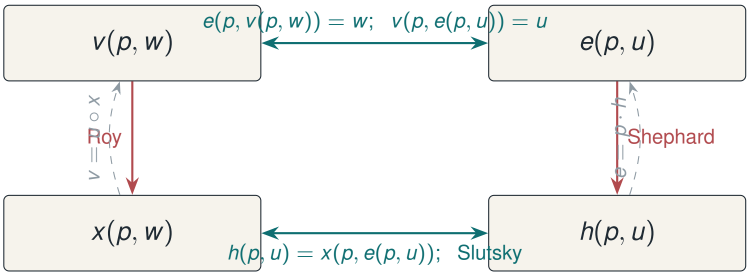
$S(p, w) = D_p x + D_w x\, x^{\mathsf{T}} = D^2_{pp} e$ is

1. **symmetric**: $s_{\ell k} = s_{k\ell}$;
2. **negative semidefinite**: $z^{\mathsf{T}} S z \leq 0$ for all z;
3. **singular with null vector** p: $Sp = 0$ (and $p^{\mathsf{T}} S = 0$).

The scorecard, completed

Lecture 1: WARP alone delivered (2) and (3), *not* (1). Symmetry is the differential fingerprint of **transitivity** — of a single coherent preference behind all choices. It is also exactly the condition for recovering preferences from demand (integrability, next lecture). One matrix, three axioms, three testable restrictions.

The square, closed



- Exam heuristic: given *any one* corner (plus regularity), you can reconstruct the other three. Prelims love starting you at v or e.

Working vocabulary for comparative statics

- **Net (Hicksian) substitutes/complements**: sign of $s_{\ell k}$. Symmetry makes this a *symmetric relation* — tea can't be a net substitute for coffee while coffee complements tea.
- **Gross (Walrasian) substitutes/complements**: sign of $\partial x_\ell / \partial p_k$; income effects break symmetry, so "gross" talk needs ordered pairs.
- Lecture 1's IOU, paid: the coherent definition of "substitutes" is the *net* one, and it cost the full rational apparatus — as promised. (One-line application: antitrust market definition is properly a net question.)
- NSD + singularity with $L = 2$: both goods are net substitutes, always ($s_{12} = -s_{11} p_1 / p_2 \geq 0$). Genuine complementarity needs $L \geq 3$.
- Elasticity form for empirical work: $\varepsilon^h_{\ell k} = \varepsilon_{\ell k} + b_k \, \eta_\ell$ with budget share b_k and wealth elasticity η_ℓ — Slutsky in the units regressions deliver.

Goal (a), executed on estimated demand

In passing: the test, run in the field

- Aguiar–Serrano (JET 2017) grade observed demand on exactly this ledger: find the nearest matrix with all three properties and measure the distance — which decomposes exactly into a symmetry part, an NSD part, and a singularity part. "Behavioral" stops being one adjective; a demand function fails *in a direction*.

Violated property	Axiomatic culprit	Example generating model
Symmetry	transitivity / integrability	sparse attention to prices
NSD	WARP-type consistency	menu-dependent choice rules
Singularity ($Sp \neq 0$)	homogeneity / Walras	money illusion

- Thursday's CCEI compresses violations into one number; this is its differential cousin that keeps the direction. Either way, the theory supplies the *coordinate system* in which its own failure is measured.

From theory to estimation

- To confront any of this with data you need $\widehat{S}$: estimate a demand system flexibly (AIDS/QUAIDS: Deaton–Muellbauer 1980; Banks–Blundell–Lewbel 1997), then form $\widehat{s}_{\ell k}$ from the estimated price and wealth derivatives.
- Classical tests impose/test symmetry and negativity as parameter restrictions; rejections in aggregate data are common — and Ch. 4's aggregation problem is one reading of why (individual rationality does not aggregate).
- Modern nonparametric route: revealed-preference bounds on finite data (Blundell–Browning–Crawford 2003 onward) — previewed properly with Afriat next lecture.

Scorecard, takeaways, readings

Takeaways

- **Goal (a): both data regimes now armed.** Finite data $\rightarrow$ GARP (you run it Thursday); demand functions $\rightarrow$ S symmetric, NSD, $Sp = 0$. Symmetry is what WARP could not deliver — transitivity's fingerprint, and (Monday) the exact condition for recovery.
- **Goal (b): tools complete.** Shephard and Roy are the derivative bridges Hausman's welfare computation walks on Monday.
- The descriptive layer, upgraded: Giffen decomposed; "substitutes" finally well-posed — net, not gross.

Required: MWG 3.F–3.G. **Next:** 3.H–3.J; finish Hausman (1981) — Monday executes it.
In passing today: Aguiar–Serrano (2017); Deaton–Muellbauer (1980); Jensen–Miller (2008).
Thursday is the practical — goal (a), run on you. Bring a laptop with R working; CKMS is the paper you are replicating.